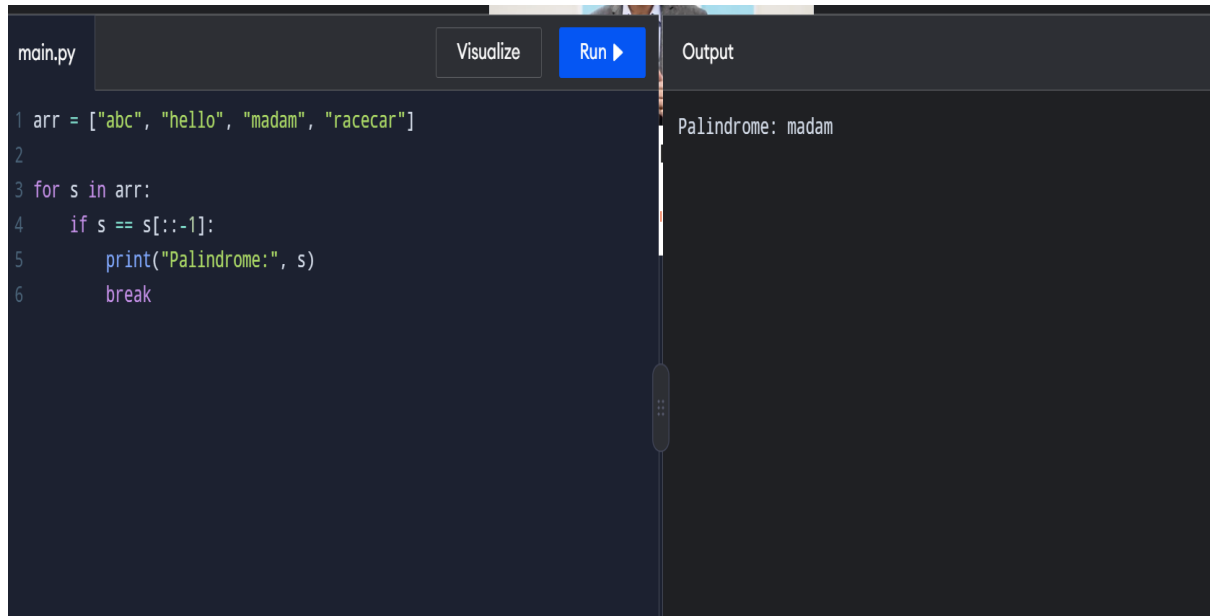


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1: First Palindromic String in an Array and find its completeness



The screenshot shows a code editor with a file named 'main.py'. The code is as follows:

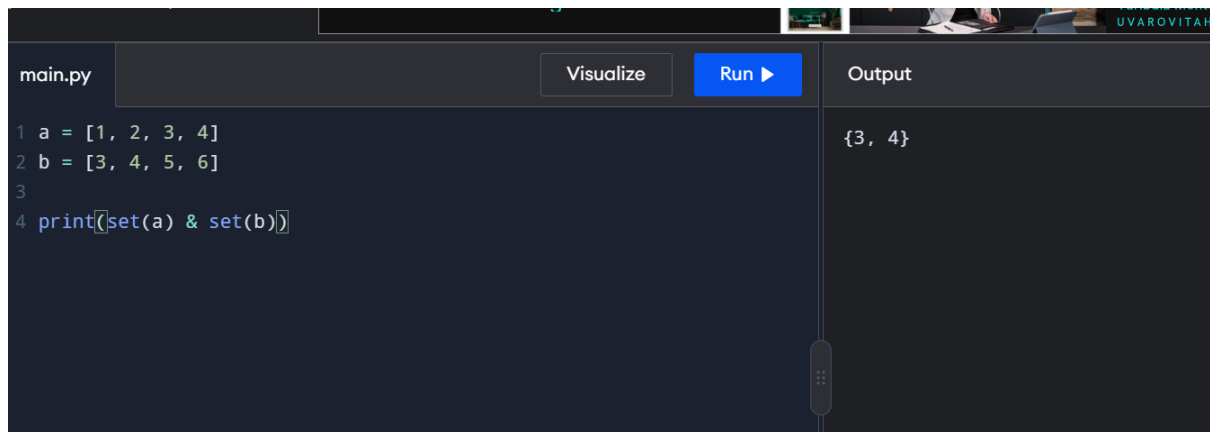
```
1 arr = ["abc", "hello", "madam", "racecar"]
2
3 for s in arr:
4     if s == s[::-1]:
5         print("Palindrome:", s)
6         break
```

Buttons for 'Visualize' and 'Run' are visible above the code. The 'Output' pane on the right displays the result: 'Palindrome: madam'.

Time: $O(n \times m)$

Space: $O(m)$

2: Common Elements Between Two Arrays and find its completeness



The screenshot shows a code editor with a file named 'main.py'. The code is as follows:

```
1 a = [1, 2, 3, 4]
2 b = [3, 4, 5, 6]
3
4 print(set(a) & set(b))
```

Buttons for 'Visualize' and 'Run' are visible above the code. The 'Output' pane on the right displays the result: '{3, 4}'.

Time: $O(n + m)$

Space: $O(n + m)$

3: Subarrays Distinct Element Sum of Squares

main.py	Visualize	Run ▶	Output
<pre>1 a = [1, 2, 1] 2 3 ans = 0 4 for i in range(len(a)): 5 s = set() 6 for j in range(i, len(a)): 7 s.add(a[j]) 8 ans += sum(x*x for x in s) 9 10 print(ans)</pre>			21

Time: $O(n^3)$

Space: $O(n)$

4: Count Equal and Divisible Pairs in an Array

main.py	Visualize	Run ▶	Output
<pre>1 a = [3, 1, 2, 3, 1] 2 k = 2 3 4 count = sum(a[i] == a[j] and (i*j) % k == 0 5 for i in range(len(a)) 6 for j in range(i+1, len(a))) 7 8 print(count)</pre>			2

Time: $O(n^2)$

Space: $O(1)$

5: Maximum Element with Least Time Complexity

main.py	Visualize	Run ▶	Output
<pre>1 a = [10, 25, 7, 40, 15] 2 print(max(a))</pre>			40

Time: $O(n)$

Space: $O(1)$